

D890

Calculus I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher writes a piecewise function for a shipping-cost example from an online store: $f(x) = x + 1$ for $x < 2$ and $f(x) = 3x - 1$ for $x \geq 2$. The store offers free returns. What is the limit of $f(x)$ as x approaches 2?

- A. 3
- B. 5
- C. The limit does not exist.
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

$f(x) = x + 1$ for $x < 2$ and $f(x) = kx - 1$ for $x \geq 2$. What value of k makes the limit at $x = 2$ exist?

- A. No value of k works.
- B. $k = 1$
- C. $k = 3$
- D. $k = 2$

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A model for average cost per unit at a print shop that opened downtown is $C(x) = (3x^2 + 5) / (6x^2 - x)$. The shop has two presses. What does $C(x)$ approach as x grows without bound?

- A. $1/2$
- B. 0
- C. -5
- D. It grows without bound.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A function f is continuous on $[1, 4]$, with $f(1) = -2$ and $f(4) = 5$. The problem appears in a workbook with a blue cover. Which statement must be true?

- A. $f(c) = 0$ for at least one c between 1 and 4.
- B. $f(c) = 0$ for exactly one c between 1 and 4.
- C. $f'(c) = 0$ for some c between 1 and 4.
- D. $f(c) = 7$ for some c between 1 and 4.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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5.

A circular fountain design in a city park is modeled by $x^2 + y^2 = 25$. The park was renovated this year and has new benches. What is dy/dx at the point $(3, 4)$?

- A. $-3/4$
- B. $3/4$
- C. $-4/3$
- D. 6

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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6.

A cart on a test track at an engineering fair has velocity $v(t) = t^2 - 6t + 8$ meters per second. The fair attracts 3,000 visitors a year. At $t = 1$ second, is the cart speeding up or slowing down?

- A. Slowing down, because negative acceleration always means slowing down.
- B. Slowing down, because velocity and acceleration have opposite signs.
- C. Speeding up, because velocity is positive.
- D. At rest, because $v(t) = 0$ at this time.

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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7. [TRANSFER — reversed stem / different variable]

For the same cart, $v(t) = t^2 - 6t + 8$. At $t = 2.5$ seconds, is it speeding up or slowing down?

- A. Slowing down, because velocity is negative.
- B. At rest
- C. Slowing down, because acceleration is negative.
- D. Speeding up

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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8.

A stone dropped in a pond at a summer camp makes a circular ripple whose radius grows at 2 cm per second. The camp has 120 campers. How fast is the area increasing when the radius is 5 cm?

- A. $50 \pi \text{ cm}^2$ per second
- B. $20 \pi \text{ cm}^2$ per second
- C. $10 \pi \text{ cm}^2$ per second
- D. $25 \pi \text{ cm}^2$ per second

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A circle's area is growing at $30 \pi \text{ cm}^2$ per second when the radius is 5 cm. How fast is the radius growing?

- A. 6 cm per second
- B. 30 cm per second
- C. 3 cm per second
- D. 1.2 cm per second

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

On a practice test used by a tutoring center, $f(x) = x^3$ and $f'(0) = 0$. The tutoring center is open on Saturdays. What does $x = 0$ represent?

- A. Not a critical point, because the function is always increasing.
- B. A local minimum, because $f'(0) = 0$.
- C. A critical point that is neither a maximum nor a minimum.
- D. A local maximum, because $f''(0) = 0$.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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11.

The temperature in a greenhouse at a botanical garden is modeled by $f(x) = x^2$ degrees above a baseline for $0 \leq x \leq 3$ hours. The garden has a gift shop. What is the average value of f on $[0, 3]$?

- A.** 9
- B.** 4.5
- C.** 1
- D.** 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

The average value of f on $[2, 6]$ is 5. What is the integral of f from 2 to 6?

- A.** 1.25
- B.** 20
- C.** 30
- D.** 10

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher writes a piecewise function ~~for a shipping cost example from an online store~~: $f(x) = x + 1$ for $x < 2$ and $f(x) = 3x - 1$ for $x \geq 2$. ~~The store offers free returns.~~ What is the limit of $f(x)$ as x approaches 2?

Target word: two-sided limit **Decides:** Left limit 3, right limit 5.

A	Dropped element	Uses only the left-hand limit.
B	Confusion set	That is $f(2)$ and the right-hand limit only.
C	Correct	The one-sided limits (3 and 5) are not equal.
D	Adjacent category	Averages the one-sided limits, which is not a rule.

2. Answer D Concept 1 · transfer

$f(x) = x + 1$ for $x < 2$ and $f(x) = kx - 1$ for $x \geq 2$. What value of k makes the limit at $x = 2$ exist?

Target word: solve for k **Decides:** Set the one-sided limits equal.

A	Absolute language	$k = 2$ matches the limits.
B	Wrong stem number	Uses $2k - 1 = 1$.
C	Dropped element	Sets k equal to the left limit.
D	Correct	Left limit 3; $2k - 1 = 3$ gives $k = 2$.

3. Answer A Concept 2

A model for average cost per unit ~~at a print shop that opened downtown~~ is $C(x) = (3x^2 + 5) / (6x^2 - x)$. ~~The shop has two presses.~~ What does $C(x)$ approach as x grows without bound?

Target word: limit at infinity **Decides:** Equal degrees: ratio of leading coefficients.

A	Correct	$3/6 = 1/2$.
B	Confusion set	That is the rule when the denominator degree is higher.
C	Wrong stem number	Uses the constant terms.
D	Confusion set	That is the rule when the numerator degree is higher.

4. Answer A Concept 3

A function f is continuous on $[1, 4]$, with $f(1) = -2$ and $f(4) = 5$. ~~The problem appears in a workbook with a blue cover.~~ Which statement must be true?

Target word: IVT conclusion **Decides:** Continuous with a sign change.

A	Correct	IVT guarantees every value between -2 and 5 is reached.
B	Absolute language	IVT guarantees at least one, not exactly one.
C	Confusion set	That is Rolle's theorem, which needs equal endpoint values and differentiability.
D	Inverted operation	7 is the difference between the values, not a value between them.

5. Answer A Concept 4

A circular fountain design in a city park is modeled by $x^2 + y^2 = 25$. ~~The park was renovated this year and has new benches.~~ What is dy/dx at the point (3, 4)?

Target word: implicit slope **Decides:** $2x + 2y y' = 0$.

A	Correct	$dy/dx = -x/y = -3/4$.
B	Inverted operation	Drops the negative when solving.
C	Confusion set	Inverts the ratio, giving $-y/x$ instead of $-x/y$.
D	Dropped element	Differentiates only x^2 and ignores the y term.

6. Answer B Concept 5

A cart on a test track at an engineering fair has velocity $v(t) = t^2 - 6t + 8$ meters per second. ~~The fair attracts 3,000 visitors a year.~~ At $t = 1$ second, is the cart speeding up or slowing down?

Target word: speeding up or slowing down **Decides:** $v(1) = 3$ and $a(1) = -4$ have opposite signs.

A	Absolute language	Right answer, wrong rule; it fails when velocity is also negative.
B	Correct	$v(1) = 3 > 0$ and $a(1) = 2(1) - 6 = -4 < 0$.
C	Dropped element	Ignores the acceleration.
D	Wrong stem number	$v = 0$ happens at $t = 2$ and $t = 4$, not $t = 1$.

7. Answer D Concept 5 · transfer

For the same cart, $v(t) = t^2 - 6t + 8$. At $t = 2.5$ seconds, is it speeding up or slowing down?

Target word: reverse case **Decides:** $v(2.5) = -0.75$ and $a(2.5) = -1$: same sign.

A	Dropped element	Ignores acceleration.
B	Wrong stem number	$v(2.5)$ is not 0.
C	Confusion set	Negative acceleration speeds up an object already moving in the negative direction.
D	Correct	Both velocity and acceleration are negative.

8. Answer B Concept 6

A stone dropped in a pond at a summer camp makes a circular ripple whose radius grows at 2 cm per second. ~~The camp has 120 campers.~~ How fast is the area increasing when the radius is 5 cm?

Target word: dA/dt **Decides:** $dA/dt = 2\pi r dr/dt$.

A	Inverted operation	Multiplies the area 25π by dr/dt instead of differentiating the area formula.
B	Correct	$2\pi(5)(2) = 20\pi$, about 62.8.
C	Dropped element	Drops $dr/dt = 2$.
D	Confusion set	That is the area, not its rate of change.

9. Answer C Concept 6 · transfer

A circle's area is growing at $30\pi \text{ cm}^2$ per second when the radius is 5 cm. How fast is the radius growing?

Target word: solve for dr/dt **Decides:** $30\pi = 2\pi(5) dr/dt$.

A	Inverted operation	Divides by 5π instead of 10π .
B	Dropped element	Drops the $2\pi r$ factor.
C	Correct	$dr/dt = 30\pi / 10\pi = 3$.
D	Confusion set	Divides 30π by the area 25π .

10. Answer C Concept 7

On a practice test used by a tutoring center, $f(x) = x^3$ and $f'(0) = 0$. ~~The tutoring center is open on Saturdays.~~ What does $x = 0$ represent?

Target word: nature of the critical point **Decides:** $f'(x) = 3x^2$ does not change sign.

A	Inverted operation	$f'(0) = 0$, so it is a critical point.
B	Absolute language	A zero derivative does not guarantee an extremum.
C	Correct	f' is positive on both sides of 0.
D	Confusion set	$f''(0) = 0$ makes the second-derivative test inconclusive.

11. Answer D Concept 8

The temperature in a greenhouse at a botanical garden is modeled by $f(x) = x^2$ degrees above a baseline for $0 \leq x \leq 3$ hours. ~~The garden has a gift shop.~~ What is the average value of f on $[0, 3]$?

Target word: average value **Decides:** $(1/3)$ times the integral of x^2 from 0 to 3.

A	Dropped element	Forgets to divide by the interval length.
B	Adjacent category	Averages $f(0)$ and $f(3)$.
C	Wrong stem number	Divides by 9 instead of by 3.
D	Correct	The integral is 9; $9/3 = 3$.

12. Answer B Concept 8 · transfer

The average value of f on $[2, 6]$ is 5. What is the integral of f from 2 to 6?

Target word: solve for the integral **Decides:** Integral = average $\times$ (b - a).

A	Inverted operation	Divides by the interval length instead of multiplying.
B	Correct	$5 \times (6 - 2) = 20$.
C	Wrong stem number	Multiplies by 6 instead of 4.
D	Wrong stem number	Multiplies by 2 instead of 4.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	7	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	One-sided limits and existence	A two-sided limit exists only when the left-hand and right-hand limits are equal; the function value at the point does not decide the limit.	Reporting $f(a)$ as the limit, or averaging unequal one-sided limits.	A piecewise function that switches rules at a point.
2	Limits at infinity	For a rational function, compare degrees: equal degrees give the ratio of leading coefficients; a bigger denominator degree gives 0.	Plugging in large numbers carelessly or saying every rational function goes to infinity.	x approaching infinity in a fraction of polynomials.
3	Intermediate Value Theorem	If f is continuous on $[a, b]$, it takes every value between $f(a)$ and $f(b)$ at least once.	Saying IVT gives exactly one solution, or confusing it with a derivative result.	A continuous function with a sign change on an interval.
4	Implicit differentiation	Differentiate both sides with respect to x and multiply every y -term's derivative by dy/dx .	Treating y as a constant.	An equation mixing x and y that is not solved for y .
5	Speeding up vs. slowing down	An object speeds up when velocity and acceleration have the same sign and slows down when they differ.	Thinking negative acceleration always means slowing down.	Velocity and acceleration values at a given time.
6	Related rates	Differentiate the relation with respect to time, then substitute the values at the instant in question.	Substituting the instant's values before differentiating, or dropping the given rate.	Two quantities changing with time and a rate asked at one moment.
7	Critical points are not always extrema	$f'(c) = 0$ marks a candidate; a local max or min requires f' to change sign (or a second-derivative test).	Assuming every critical point is a maximum or minimum.	A derivative that is zero but does not change sign, such as $3x^2$.
8	Average value of a function	The average value of f on $[a, b]$ is $(1 / (b - a))$ times the integral of f from a to b .	Forgetting to divide by the interval length, or averaging the endpoint values.	The phrase average value over an interval.

Course scope (public description): Calculus I offers a foundational exploration of the essential concepts and applications of calculus, crucial for higher-level mathematics and various scientific fields. This course begins with an overview of calculating limits and continuity of functions, setting the stage for the study of derivatives. Then, students will explore applications of derivatives, including applications to objects in motion, optimization, and related rates. Finally, the course focuses on both definite and indefinite integrals, leading to the mastery of the Fundamental Theorem of Calculus. This course prepares students for advanced mathematical studies and provides the groundwork for advanced calculus topics addressed in Calculus II and Multivariable Calculus.